

Feedback — How Do We Find Disease-Causing Mutations?

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Thank you. Your submission for this quiz was received.

You submitted this quiz on **Sat 4 Apr 2015 1:03 PM PDT**. You got a score of **20.00** out of **20.00**.

Question 1

True or False: The most efficient way to map reads to a reference genome is by assembling the reads into a genome, then comparing this genome with the reference.

Your Answer	Score	Explanation
☐ True		
☒ False	✓ 2.00	
Total	2.00 / 2.00	

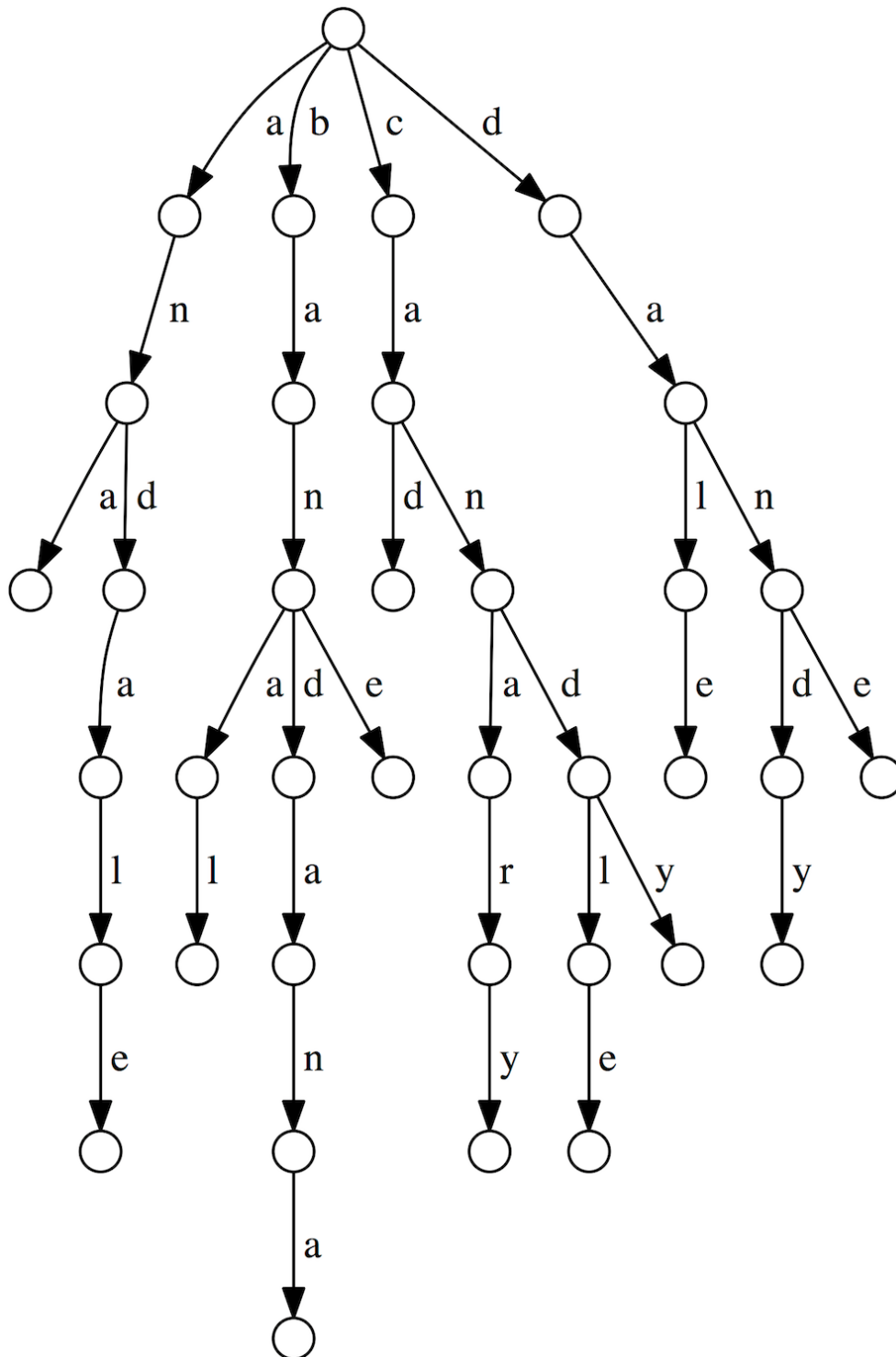
Question 2

What is the runtime of BruteForcePatternMatching?

Your Answer	Score	Explanation
☒ $O(Text * Patterns)$	✓ 2.00	
☐ $O(Text ^2)$		
☐ $O(LongestPattern)$		
☐ $O(Text * Patterns + Patterns)$		
Total	2.00 / 2.00	

Question 3

Which patterns could have been used to construct the following trie? (Select all that apply.)



Your Answer

Score

Explanation

☐ andy



0.50

☐ ananas



0.50

☒ ana



0.50

cad



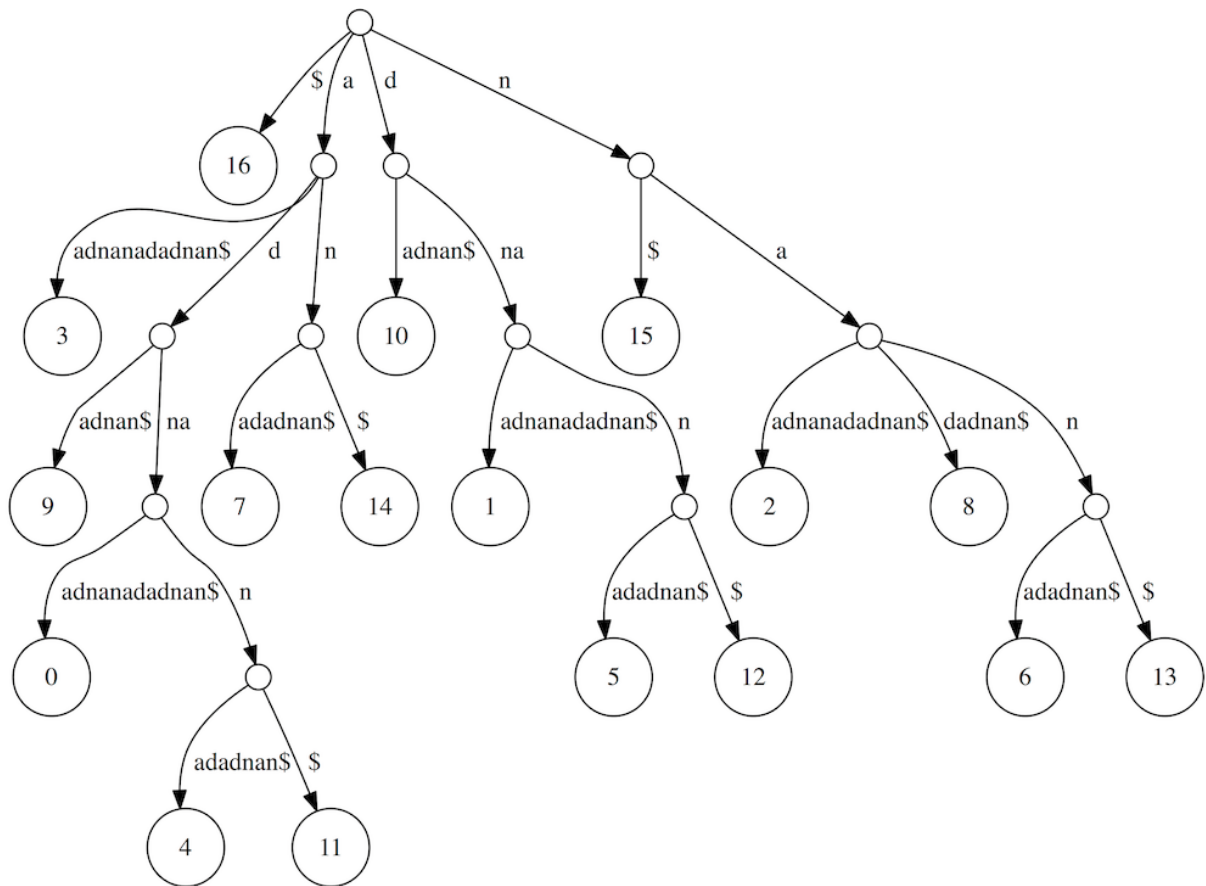
0.50

Total

2.00 / 2.00

Question 4

Given the following suffix trie/suffix tree of an unknown text, identify all starting positions in the underlying text of Pattern = "an". Use 0-based indexing and enter the starting positions as space-separated integers **in increasing order**.



You entered:

7 14

Your Answer

Score

Explanation

7



1.00

14



1.00

Total

2.00 / 2.00

Question 5

How many leaves will

SuffixTree("TCTGAGCCCTACTGTCTGAGAAATATGTATCTCGCCCCCGCAGCTT\$") have?

You entered:

46

Your Answer		Score	Explanation
46	✓	2.00	
Total		2.00 / 2.00	

Question 6

True or False: A key feature of the Burrows-Wheeler Transform is that it transforms runs into repeats.

Your Answer		Score	Explanation
☐ True			
☒ False	✓	2.00	
Total		2.00 / 2.00	

Question 7

Find the Burrows-Wheeler transform of *Text* = ACCAACACTG\$.

You entered:

GCA\$CCAAATC

Your Answer		Score	Explanation
GCA\$CCAAATC	✓	2.00	

Total

2.00 / 2.00

Question 8

If $BWT(Text) = TTCCATTGGA\$$, what is $Text$? (Don't forget to add the "\$" to the end of $Text...$)

You entered:

Your Answer		Score	Explanation
TGTACCATGT\$	✓	2.00	
Total		2.00 / 2.00	

Question 9

Which of the following structures did we use in this chapter to decrease memory when solving the Multiple Pattern Matching Problem with the Burrows-Wheeler transform? (Select all that apply.)

Your Answer		Score	Explanation
☒ partial suffix arrays	✓	0.50	
☐ de Bruijn graphs	✓	0.50	
☐ skew diagrams	✓	0.50	
☒ checkpoint arrays	✓	0.50	
Total		2.00 / 2.00	

Question 10

Say that you know that two strings of length 363 match with at most 5 mismatches, but you don't know what the strings are. What is the largest value of k such that we can guarantee that the two

strings share a k -mer?

You entered:

60

Your Answer		Score	Explanation
60	✓	2.00	
Total		2.00 / 2.00	